# Statistical & Theoretical Assumptions

BrainKAN Interpretability Benchmark

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This document details the formal mathematical and statistical assumptions underlying the BrainKAN Interpretability Benchmark, defining the conditions required for valid non-linear interpretability.

## Assumption 1: Exchangeability of Label Permutations

**Statement:** Under the null hypothesis $H_0$ (no task-associated non-linear structure), the distribution of the geometric test statistic $G(f, X)$ is invariant under the permutation of task labels $Y$:

$$(X_i, Y_i)_{i=1}^n \stackrel{d}{=} (X_i, \pi(Y)_i)_{i=1}^n \tag{1}$$

**Condition:** Labels are randomized globally at the subject level via paired sign-flips to preserve subject-level graph correlation structures.

## Assumption 2: Null Preserves Nuisance Geometry

**Statement:** The label-shuffled null model $f_{\text{null}}$ successfully captures the baseline geometric deformation induced by model expressivity and input feature routing:

$$G(f_{\text{null}}, X) = D(X, f_{\text{null}}) \tag{2}$$

**Empirical Validation:** BrainKAN models demonstrate an invariant baseline curvature budget of $T \approx 0.179 \pm 0.001$ across both real and null labels, confirming that the network inherently consumes a fixed non-linear capacity budget.

## Assumption 3: Common Support Evaluation Domain

**Statement:** Spline evaluations must be restricted strictly to the common support between task classes:

$$\mathcal{S}_{\text{common}} = \text{supp}(X_0) \cap \text{supp}(X_1) \tag{3}$$

**Violation Consequence:** As demonstrated in Exp 2-B, evaluating outside the common support causes artificial DFL inflation up to $+0.1195$, invalidating any biological interpretation.

## Assumption 4: Optimization Marginalization (SMPT)

**Statement:** Valid population-level inference regarding task-associated function properties requires marginalizing over the optimization distribution $\mathcal{S}$:

$$\Delta_{\text{marginal}} = \mathbb{E}_{s \sim \mathcal{S}} \left[ G(f_{\text{real}}^{(s)}, X) - \mathbb{E}_{\pi}[G(f_{\text{null}}^{(s)}, X)] \right] \tag{4}$$

**Violation Consequence:** Conditioning on a single initialization seed yields conditional significance (e.g., Seed 42, $p = 0.0060$) that fails to generalize across independent initializations (Exp 5: $\bar{\Delta} = -0.00058$, $p = 0.9513$).